Infertility Treatment Trend Analysis (2020-2025)

1. Project Overview and objectives

- The infertility Treatment Management System is a data management Project designed to analyses the number of infertility cases has increased over the past 5 years.
- The main objective of the project is **to design a structured database** for storing infertility patient records
- To maintain accurate patient information and to track different infertility types and primary causes affecting patients
- To calculate and analyses the total amount spent by patient on infertility treatments for financial assessment and reporting.

2. Data source:

The data used in the city-wise infertility Treatment Trend analysis (2020-2025). The dataset which I chose is synthetically generated dataset created for academic analytical purposes. The structure and parameters are designed based on publicly available infertility statistics.

3. Problem Statement

- Infertility has become the one of the major problems now a days which goes medical and financial concern over the past 5 years.
- There is an increasing number of infertility cases along with rising treatment costs such as IVF, IUI, Medication etc.
- Many hospitals lack a structured system to analysis the treatment trends, success rates and financial burden city-wise

  The absence of organized data makes it difficult to

  i. Compare infertility cases across different cities.
  ii. Analysis treatment success rates
  iii. Evaluate the total amount spent by infertility
  iv. Causes of infertility
  v. Study the impact of insurance coverage
  vi. Identify cost trends over time.

4. Attribute:

Attribute Name	Datatype	Description in table
Patient_ID	Text	Unique identifier assigned to each patient
Gender	Text	Specifies the gender of the patient
Age	Whole number	Age of the patient in years
Hospital	Text	Name of the hospital where patient receives treatment
Infertility_Type	Text	Denotes the infertility type (primary or secondary)
Primary_Cause	Text	Reason Causing infertility
Treatment_Type	Text	Patient undergoes the Type of treatment (IUF, IVF, medication or Surgery)
Treatment_Cost_INR	Currency	Cost of the treatment in INR
Treatment_Cycles	Whole number	Number of cycles the patient has undergone success.
Success	Text	Indicates whether the treatment was success
Insurance_Coverage	Text	Insurance covers the treatment
Year	Whole number	Year in which the treatment was undertaken

5. Tools and Technology:

Excel: Excel is used for preprocessing and Pivot tables

Power BI: Power BI is used for Visualization and DAX calculations

6. Data Preprocessing using Excel:

Imported data's from csv file to Excel and in power Query editor.

- Changed the datatypes for all fields
- Removed the duplicates
- Standardized the formats

PIVOT TABLES:

Pivot table1: City Vs Treatment Cost

CITY	Sum of Treatment_Cost_INR
Bangalore	$22,662,905.00
Chennai	$22,281,466.00
Delhi	$19,482,774.00
Hyderabad	$22,788,155.00
Kolkata	$25,122,890.00
Mumbai	$21,984,361.00
Pune	$26,855,663.00
Grand Total	**$161,178,214.00**

Pivot table-2: Hospital Vs treatment cost

Row Labels	Sum of Treatment_Cost_INR
AMRI Hospital	$25,122,890.00
Apollo Hospital	$22,281,466.00
Fortis Hospital	$21,984,361.00
Manipal Hospital	$22,662,905.00
Max Hospital	$19,482,774.00
Ruby Hall Clinic	$26,855,663.00
Yashoda Hospital	$22,788,155.00
Grand Total	**$161,178,214.00**

PIVOT Table 3: Count of Patient based on the treatment Type:

Treatment T	Count of Patient_ID
IUI	292
IVF	317
Medication	314
Surgery	277
Grand Total	**1200**

Pivot table 4: Age Vs Count of Success rate

Patient Age	Count of Success rate
24	59
25	52
26	59
27	49
28	55
29	55
30	52
31	60
32	56
33	52
34	55
35	55
36	56
37	47
38	61
39	65
40	55
41	57
42	57
43	43
44	46
45	54
Grand Total	**1200**

PIVOT Table 5:
Year-wise Treatment Cost:

Year	Sum of Treatment_Cost_INR
2020	28829643
2021	26558176
2022	26545303
2023	27762672
2024	24764064
2025	26718356
Grand Total	**161178214**

PIVOT Table 6:

Primary cause of Fertility Based on Age of Gender:

Fertility Primary Cause	Average of Age
Female	
Advanced Maternal Age	34
Blocked Fallopian Tubes	35
Endometriosis	34
Ovulation Disorder	34
PCOS	34
Unexplained Infertility	34
Male	
Genetic Factors	34
Hormonal Imbalance	35
Infection	35
Low Motility	34
Low Sperm Count	35
Obstruction	35
Varicocele	34
Grand Total	**34**

PIVOT Table: 7

Treatment cycles Vs Success Rate:

Treatement Cycles	Count of Success
1	592
2	446
3	162
Grand Total	**1200**

PowerBI:

PowerBI is used to build a report/ dashboard with Multiple visualization Like

- Analysis the treatment Success rates
- Compare the Primary causes of infertility
- Identify the cost trend across the hospital in different cities
- Analysis the infertility happened based on their age and gender
- Analysis of Primary Causes of Infertility by Average Age:
- Success Count by treatment cycles.

7. Data modeling and DAX:

Total Count:

```
Total_Patient = Count(infertility_Data[Patient_ID])
```

Total Revenue:

```
Total Revenue = sum('CITY VS Treatment Cost'[Sum of
                Treatment_Cost_INR])
```

Success Count:

```
Success Count = CALCULATE(Count('infertility_Data'[Patient_ID]),
                infertility_Data[Success]="Yes")
```

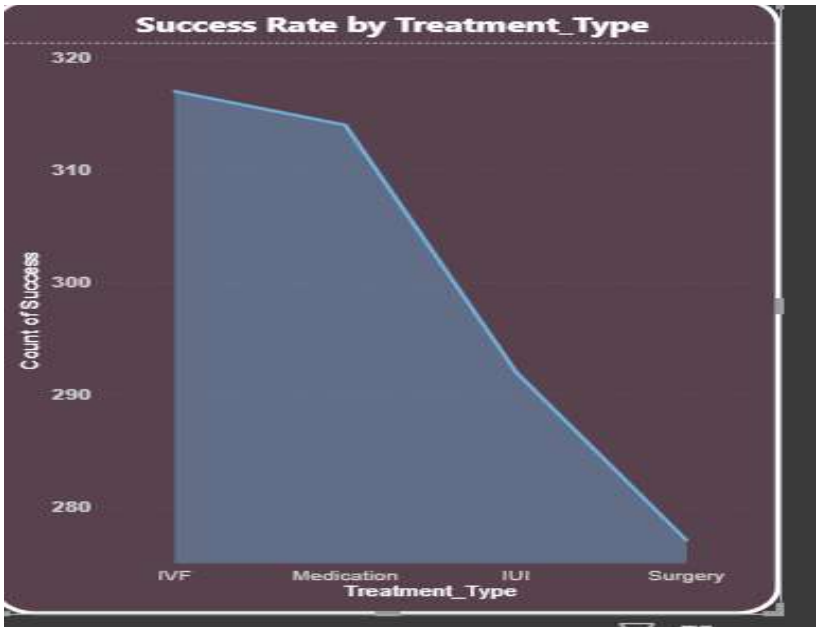
Gauge :

Create measures for `Max Value= 1, Min value= 0,  Target value = 0.7`

```
Success_Rate% = Divide([Success Count], [Total_Patient])
```

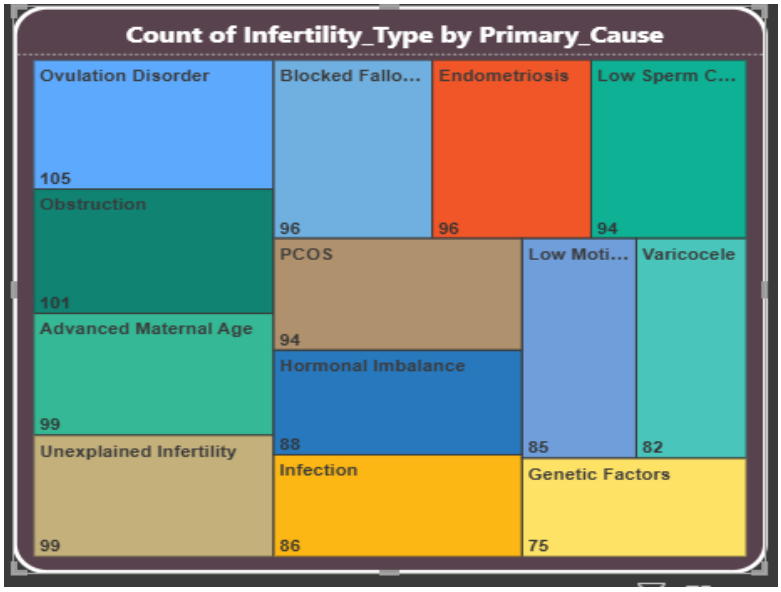
8. Analysis and Visualization: (PowerBI)

- **Analysis the treatment Success rates**



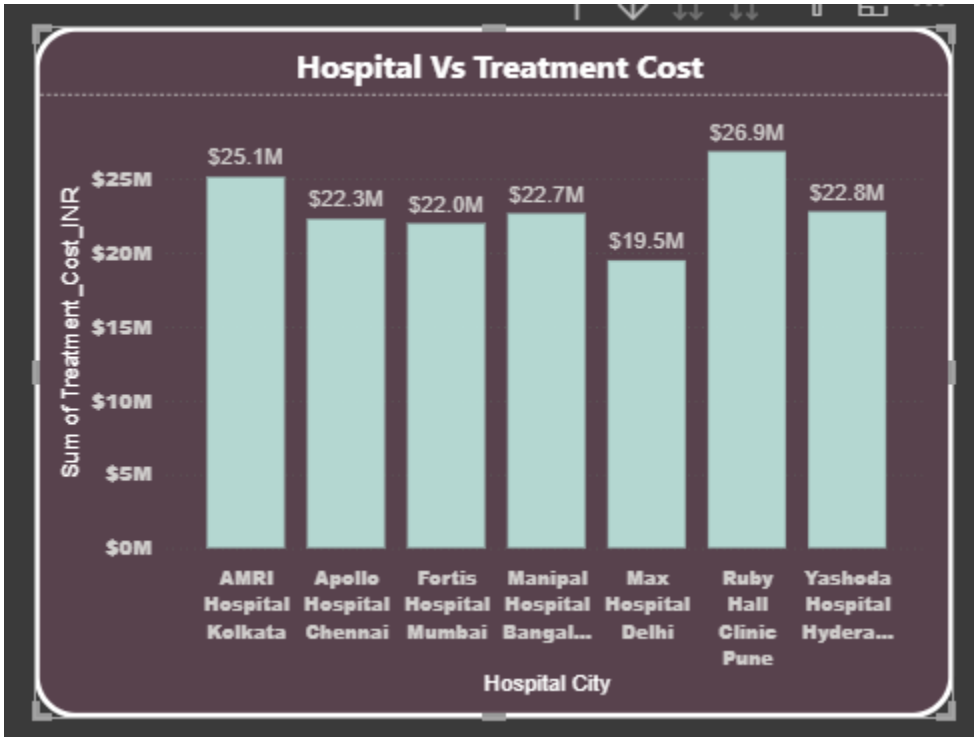
Insights: Based on this we came to know that IVF treatment gives the best success rate when compare to other, Followed by medication, IUI and Surgery

- **Compare the Primary causes of infertility**



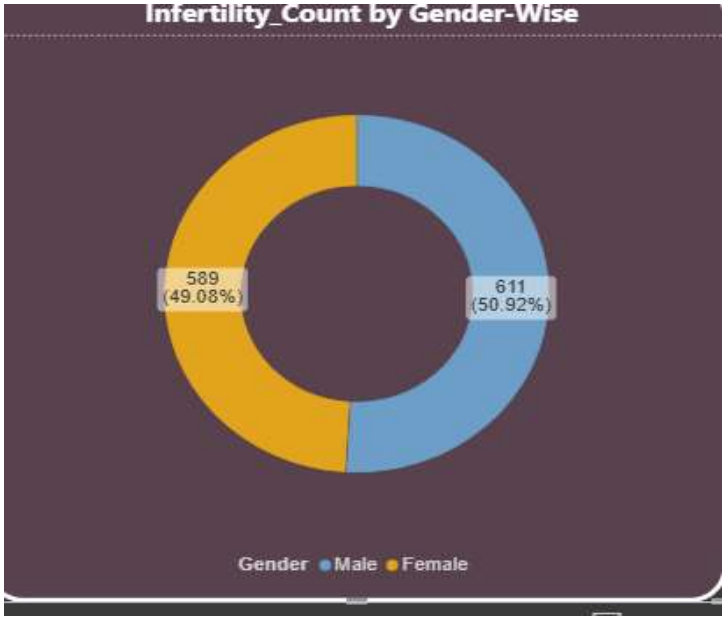
Insights: Showed this primary cause using the tree map. Based on this chart we came to know the top most primary cause of infertility patient affected by the ovulation disorder and the least is Genetic factors.

- **Identify the cost trend across the hospital in different cities**



Insights: According to the chart, Ruby Clinic in Pune has the highest costs, while Max Hospital in Delhi charges the least

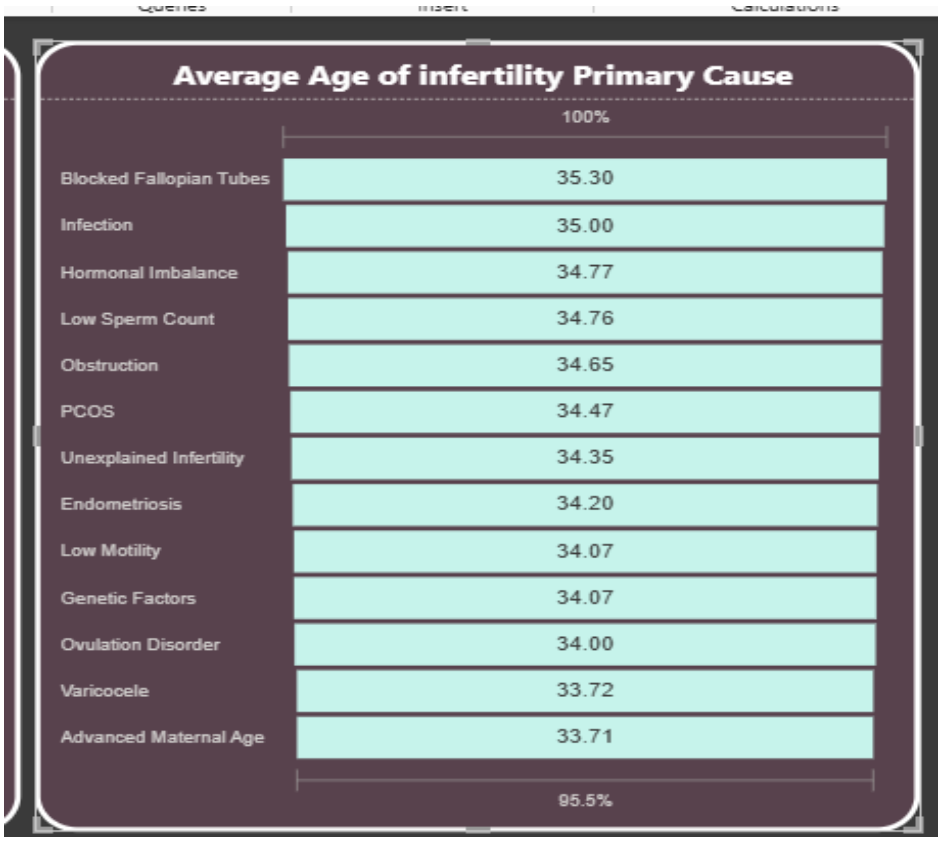
- **Analysis the infertility happened based on their age and gender**



Insights:

- Male patients account for 50% slightly higher than the female patients at 49%
- The gender distribution is almost equal indicating infertility affects both males and females at similar rate.
- The small percentage difference suggests infertility is not heavily skewed towards one gender in the dataset.

➢ **Analysis of Primary Causes of Infertility by Average Age:**



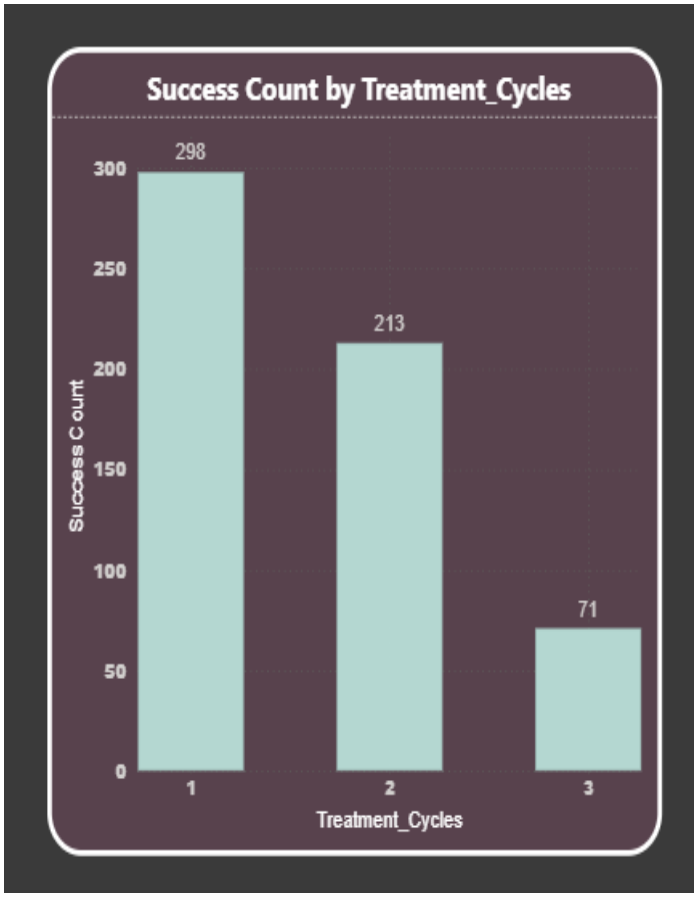
Average Age of infertility Primary Cause

Blocked Fallopian Tubes	35.30
Infection	35.00
Hormonal Imbalance	34.77
Low Sperm Count	34.76
Obstruction	34.65
PCOS	34.47
Unexplained Infertility	34.35
Endometriosis	34.20
Low Motility	34.07
Genetic Factors	34.07
Ovulation Disorder	34.00
Varicocele	33.72
Advanced Maternal Age	33.71

Insights:

- Infertility Cases are mainly concentrated in tha age group 34-35 years, which indicates higher risk at the mid of 30's.
- Structural causes such as Blocked Fallopian Tubes and infection shows slightly higher average age compare to other factors

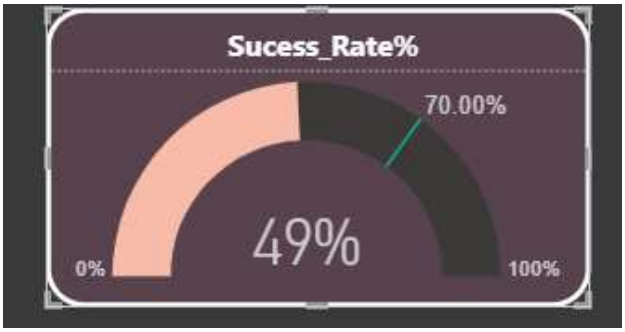
- **Success Count by treatment cycles.**



Insights:

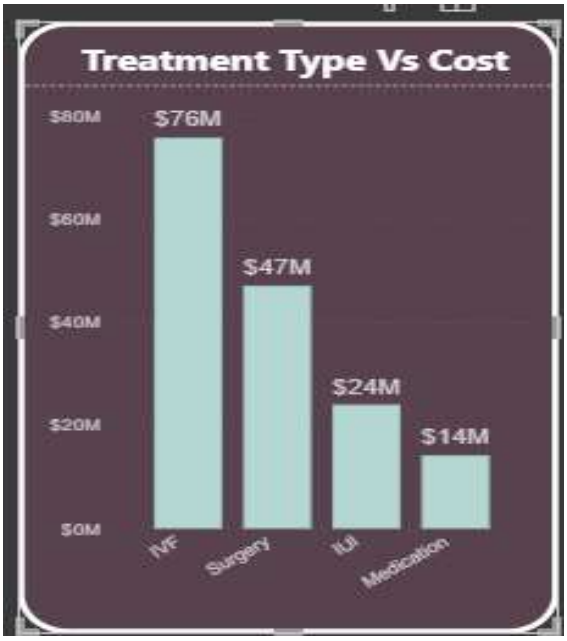
- Treatment cycle 1 shows the highest success count compared to other cycles.
- This indicated most of the patients achieve successful outcomes during their first attempt.
- Followed by treatment cycle 2 and 3, which shows comparatively lower

- **Gauge**



Insight: The Current success rate is 49%, which is 21% below the 70% target. Improvement in treatment effectiveness is needed to achieve the desired performance level.

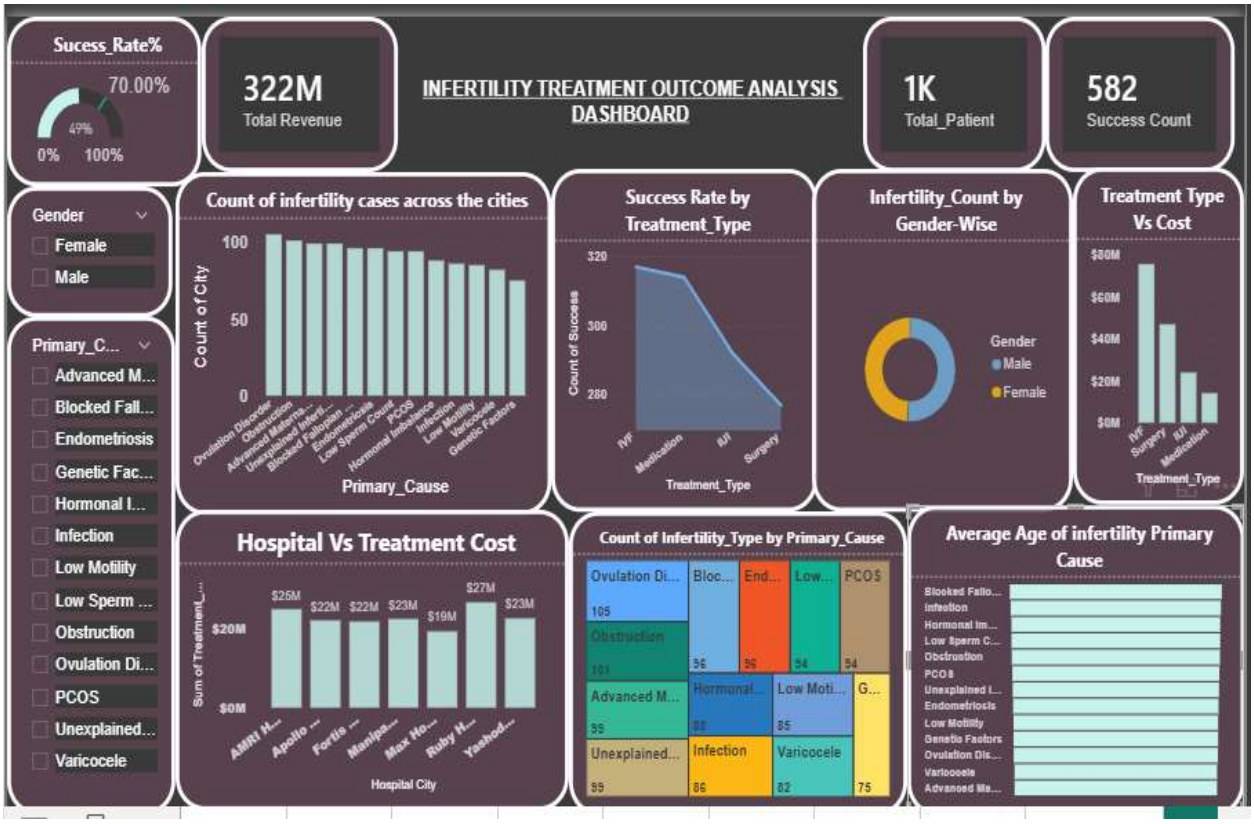
- **Treatment Type Vs Cost**



Insights:

According to cost wise of treatment (IVF) shows the highest revenue when compare to other treatment types

9. Dashboard:



10. Analysis insight (Types of Analysis)

Descriptive analysis:

In this infertility project, descriptive analysis was used to:

- ✓ Calculate total number of patient.
- ✓ Determine overall success rate
- ✓ Measures total treatment Revenue.
- ✓ Identify average patient age
- ✓ Analysis the distribution of treatment type and cost of the treatment type across the cities

This analysis provides a clear summary of the dataset and helps understand the overall performance of infertility treatments.

Diagnostic analysis:

Diagnostic analysis identifies reasons behind trends and patterns found in descriptive analysis

In this project, diagnostic analysis was used to:

- ✓ Examine why certain age groups show higher success rates.
- ✓ Analyze why IVF treatment generates higher revenue
- ✓ Compare success rates across treatment cycles.
- ✓ Identify primary causes contributing to infertility.

This analysis helps explain the factors influencing treatment Outcomes and financial performance.

Predictive analysis:

- ✓ The analysis indicates that the patients in the age group 25-30 tend to show higher treatment success compared to older age group.
- ✓ Based on this pattern, it can be predicted that younger patients are more likely to achieve successful outcomes on future treatment cycles.
- ✓ Similarly, the data shows that certain treatment, such as IVF, demonstrate comparatively higher success rates. Therefore, it can be projected that IVF may continue to remain the most effective and revenue- generated treatment option.

Prescriptive analysis:

- ✓ Hospital can focus on high-success treatments like IVF
- ✓ Early treatment is recommended for better success rates.
- ✓ Cost optimization strategies can be applied in high-expense cities.
- ✓ Awareness programs can target common infertility causes

11. Conclusion:

This project analyzed infertility treatment data using Excel and Power BI to generate meaningful insights. Through data preprocessing, interactive dashboard, key metrics such as patient distribution, success rates and treatment costs were examined. The analysis identified patterns across age groups, treatment types and cities. Overall, the project demonstrates how data analytics can support informed decision-making in healthcare management.